Small-Sample (*t-tools*) Tests

Applications: $\mu,\ \mu_1 - \mu_2$

Suppose $\{X_1,...,X_n\}$ is a random sample of independent and identically distributed $Normal(\mu_1, \sigma_1^2)$ random variables, and $\{Y_1,...,Y_n\}$ is a random sample of independent and identically distributed $Normal(\mu_2, \sigma_2^2)$ random variables. For the tests described below:

a) For fixed α, what is k in the RR?

b) Find the p-value of the test.

1. Consider testing $H_0 : \mu_1 = \mu_0 \ vs. \ H_a : \mu_1 > \mu_0$, where σ_1^2 is an unknown constant.

   *Repeat for the left-tailed and two-tailed tests.

2. Consider testing $H_0 : \mu_1 - \mu_2 = \mu_0 \ vs. \ H_a : \mu_1 - \mu_2 > \mu_0$, where σ_1^2 and σ_2^2 are unknown constants, but it is assumed $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

   *Repeat for the left-tailed and two-tailed tests.

Tests for Variances

Applications: σ^2, σ_2^2/σ_1^2

Suppose $\{X_1,...,X_n\}$ is a random sample of independent and identically distributed $Normal(\mu_1,\sigma_1^2)$ random variables, and $\{Y_1,...,Y_n\}$ is a random sample of independent and identically distributed $Normal(\mu_2,\sigma_2^2)$ random variables. For the tests described below:

a) For fixed α, what is k in the RR?

b) Find the p-value of the test.

1. Consider testing $H_0:\sigma^2=\sigma_0^2 \; vs. \; H_a:\sigma^2>\sigma_0^2$.

   Define $RR=\{S^2 : s^2>k\}$.

   *Repeat for the left-tailed and two-tailed tests.

2. Consider testing $H_0:\sigma_1^2/\sigma_2^2=1 \; vs. \; H_a:\sigma_1^2/\sigma_2^2>1$.

   Define $RR=\{(S_1^2,S_2^2) : s_1^2/s_2^2>k\}$.

   *Repeat for the left-tailed and two-tailed tests.